Colour Changing Solution by shaking

Equipment

- Flask
- warm water
- Stopper
- glucose solution
- sodium hydroxide
- methylene blue dye.

Method

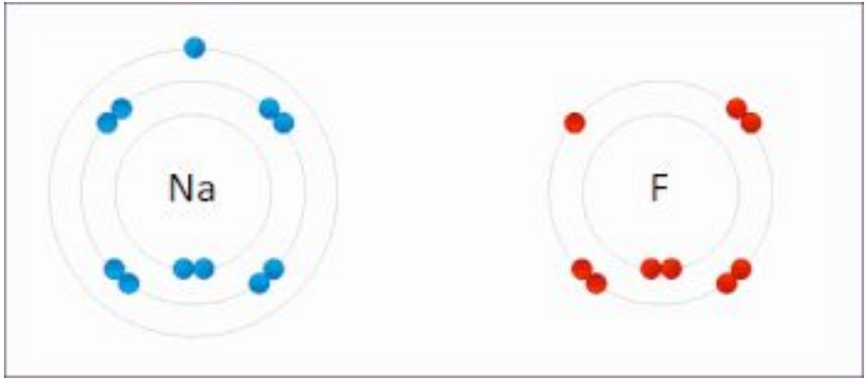
1. Add about 50 ml of warm water onto the flask.
2. Add 3 drops of methylene blue dye.
3. Add about 10 ml of glucose solution.
4. Add about 5 ml of Sodium hydroxide solution.
5. Place the stopper onto the flask, so no liquid can escape the flask.
6. Wait 10 seconds.
7. Shake the flask, holding the stopper with your finger.

Explanation

What is happening? Why?

- When the dye reacts with the glucose, it loses its colour.
- When the solution is shaken, more **oxygen** dissolves in the solution.
- The **oxygen** reacts again with the **dye**, returning its colour
- In this reaction, the colour change is due to the **reduction** of **methylene blue** by **glucose** in the presence of **sodium hydroxide**. Initially, the blue colour of methylene blue is **reduced** to a **colourless** form by **glucose**. When you **shake** the flask, introducing **oxygen**, the methylene blue is **oxidized** back to its **blue** form. This process involves the interplay of oxidation and reduction reactions.
- The colour change in oxidation-reduction reactions is associated with changes in the electronic structure of molecules. In the case of methylene blue, the colour is influenced by the **oxidation state** of the molecule.
- When methylene blue is reduced, it gains electrons, leading to a colour change from blue to colorless. This alteration in the **electronic configuration** affects the **absorption of light** by the molecule, resulting in a **different color appearance**. Conversely, when methylene blue is **oxidized**, it **loses electrons**, and the molecule reverts to its **original blue colour**.

Oxidation
Is
Loss
Reduction
Is
Gain



What else could we do to make the dye react with more **oxygen** and return it to its **original colour**?